1. Evidence review: the isogeny-chain factor base under an F4 solver

Decision DEC-20261015-ab6f23: **reject_scoped**, scope widened to msolve · **evidence** EV-ECDLP-7e970a (contradicts, replicated, toy tier) · **validation** VAL-20261015-280100 (passed) · **run** RUN-ECDLP-502ae2-01 · **experiment** EXP-ECDLP-502ae2 · **predecessor** EXP-ECDLP-f49f75 (DEC-20261015-a08389, Singular) · **goal** GOAL-ECDLP-001.

Scoped claim. On the same $m = 3$ instances as EXP-ECDLP-f49f75 (p 10–22 bits), msolve 0.6.5 (F4 + FGLM) solves the 2-isogeny-chain encoding of the factor-base set more slowly than the dense encoding of the same set and no faster than a random set, and the chain arm is retired first. The Singular-only rejection of H-ECDLP-60de36 extends to this F4 solver.

1.1. Question

DEC-20261015-a08389 left one obvious objection open: Singular's `std`/`slimgb` are Buchberger-type algorithms, and the chain's many low-degree equations are exactly the structure F4-style linear algebra is meant to exploit. This experiment re-ran the identical $m = 3$ instances under msolve.

1.2. Observation

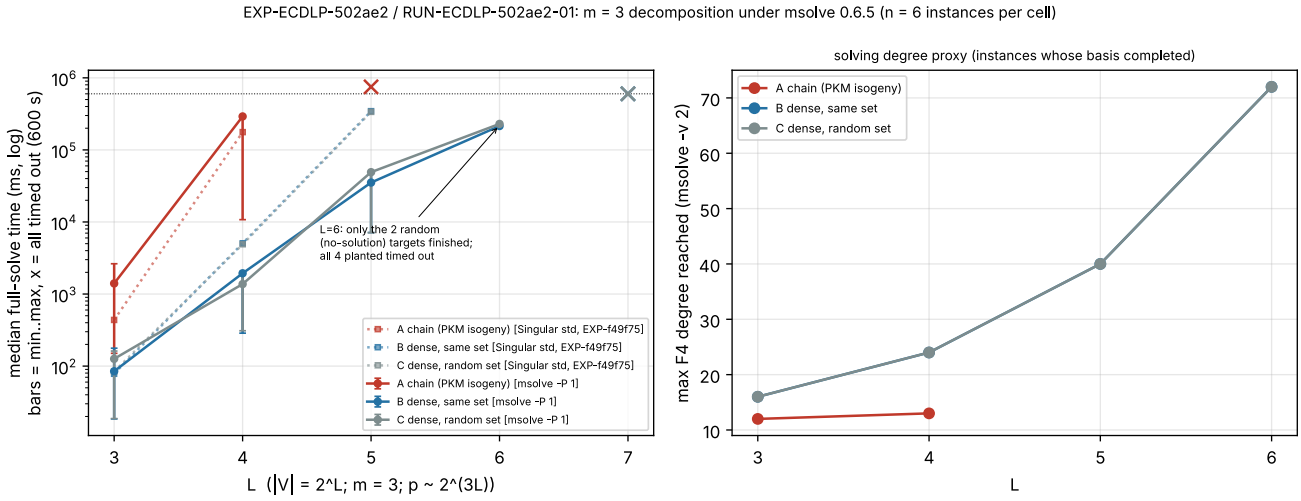


Figure 1: Median full-solve time (msolve `-P 1`, measured) vs L; dotted: Singular on the same instances; × = all instances timed out at 600 s. Right: maximum F4 degree reached. n = 6 per cell.

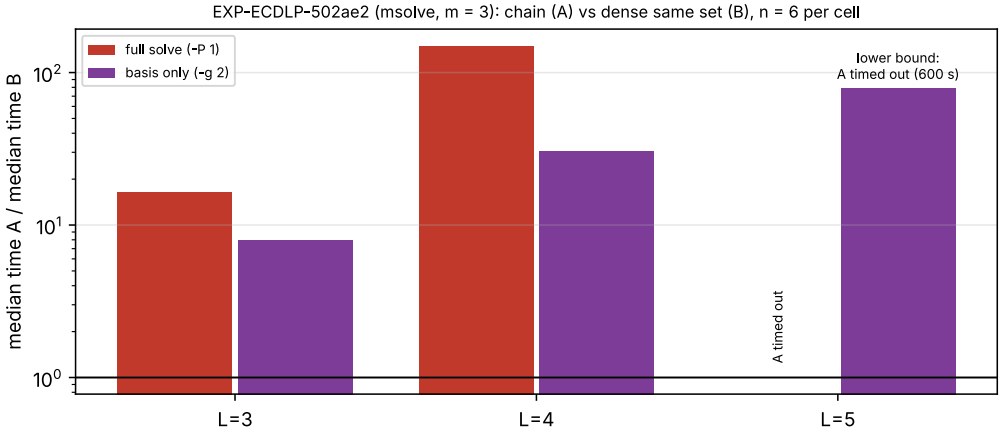


Figure 2: A/B median-time ratio (derived from measured medians, `summary.json`). The basis-only ratio isolates F4 from msolve's `-P 1` genericity retries, which inflate A's full-solve time about 5× (validator check 4c). At $L = 5$ A's basis step timed out on 6/6, so the bar is a lower bound.

frozen criterion	threshold	observed
A faster than B in jointly completed cells	≥ 80% supports; A ≥ B in ≥ 80% falsifies	0/2 faster (L = 3: 16.5×, L = 4: 150× slower)
retirement	A must complete a larger L than B and C	A retired L = 5; B, C retired L = 7

positive control	100%	32/32 planted points satisfy msolve's reduced basis

1.3. Comparison

- **Robust to the instrument's solve step:** basis-only ratios are 8.0×, 30.5× and $> 79\times$ at $L = 3, 4, 5$.
- **Lower degree, higher cost:** the chain keeps the maximum F4 degree at 12–13, while the dense arms climb $16 \to 24 \to 40 \to 72$ ($\approx$ the univariate degree 2^L). With $3L$ variables the F4 matrices still grow faster than the degree saving helps.
- **No structure seen:** B (isogeny set) and C (random set) cost the same; enumeration needs 629 group operations at $L = 5$.
- **Across instruments:** msolve is faster than Singular on the dense arms from $L = 4$ (Singular/msolve $\approx$ 2.8–11× at $L = 4, 5$ for `std`) but not on the chain arm ($\approx$ 0.2–0.8×). The secondary ratio divides Singular's internal timer by msolve wall time and is indicative only.

1.4. Inference

Moving from Buchberger to F4 does not rescue the chain encoding. The question that DEC-20261015-a08389 left open — instrument dependence — is answered negatively for the obvious F4 solver. The decomposition step needed for the proposed $N^{\frac{1}{3}}$ P-256 route remains without any toy-scale signal.

1.5. Limitations

- $m = 3$ only, toy primes, msolve 0.6.5 single-threaded, DRL in the launcher's variable order, `-c 2`, 600 s per step. Block or elimination orders that exploit the chain's triangular shape, other F4/F5 implementations and $m \geq 4$ were not tested. msolve 0.6.5 crashes on the $m = 2$, $L = 8$ dense system, so $m = 2$ was not re-run.
- At $L = 6$ the dense arms completed only the two random (empty-variety) targets; counting planted instances only, B and C reach $L = 5$ against A's $L = 4$ — same verdict.
- `-P 1` over tiny fields sometimes lists an incomplete solution set (both arms, random); "solve ok" means msolve finished. No peak-memory record in the run (validator measured 252 MB for A, 20 MB for B at $L = 4$).

1.6. Next actions

1. Problem record stays open; the F4 unblock path is tested and negative. Remaining routes: a block order eliminating the chain variables first, and an L-independent degree-of-regularity argument.
2. No further P-256 work on this route unless one of those beats the dense and random controls on this frozen grid.